DSA REFERENCE

Topological Sort

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Order a directed acyclic graph so every prerequisite appears before what depends on it.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

DSA / TOPOLOGICAL-SORT

DSA

Topological Sort

CORE IDEA

Order a directed acyclic graph so every prerequisite appears before what depends on it.

Recognition cues

+ prerequisites
+ build order
+ dependency scheduling
+ detect directed cycle

Implementation sequence

+ build adjacency and indegree
+ seed all zero-indegree nodes
+ decrement each outgoing edge once
+ compare output count with node count

Invariant

Every queued node currently has zero unresolved incoming dependencies.

Worked reasoning

Course Schedule succeeds when Kahn's algorithm emits all courses; a shorter output proves a cycle.

Complexity

O(V + E) time and space.

Failure modes

- [ ] reversing edge direction
- [ ] missing isolated nodes
- [ ] duplicate-edge indegree errors
- [ ] using undirected cycle logic

Practice ladder

Topological Sort: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Course Schedule**	State the cue, invariant, edge case, and proof before implementation.
2	**Course Schedule II**	State the cue, invariant, edge case, and proof before implementation.
3	**Alien Dictionary**	State the cue, invariant, edge case, and proof before implementation.
4	**Minimum Height Trees**	State the cue, invariant, edge case, and proof before implementation.
5	**Parallel Courses**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS
REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.